MBTI-Fashion Vibe Network

Living Room Anchor & Compromise Bridge for Newlyweds

MADT7202 - Web & Social Network Analytics (Midterm) | Kanyathat Rakdee | 6810424002

Business question. When two newlyweds have different MBTI labels and no shared direct Fashion Vibe, which style could provide a useful starting point for a shared living-room direction?

DATA & TRACEABILITY

67	268	16 x 12	SHA-256
raw images checked against repo listing	evidence rows (4 per image ID)	MBTI labels x Fashion Vibes	source CSV checksum recorded at run time

The analysis reads **data/raw_image_268_rows.csv** as the source of truth, checks four rows per image ID and the presence of all 67 raw filenames in the repository image directory, and writes an image-mapping audit manifest. The manifest verifies repository file presence and the CSV-declared mapping; proxy-to-pixel correspondence is not independently verified.

METHOD

Network construction	Measures	Robustness
Bipartite graph: 16 MBTI nodes and 12 researcher-defined Fashion Vibe nodes. Edge weights use image-level NPMI; retain positive-NPMI edges. Distance = 1 / NPMI.	Degree, weighted/unweighted betweenness, direct MBTI-pair coverage, fractional Bridge Load over weighted shortest paths, and shortest-path mediation.	Leave-one-creator-out tests for all six creators and minimum co-occurrence thresholds of 1, 2, and 3. Threshold outputs include active-node counts, components, and unreachable pairs.

BASELINE RESULTS

Finding	Recalculated result	Interpretation
Living Room Anchor candidate	Clean-Traditional: degree 8; direct pair coverage 28 / 120 (23.3%); weighted betweenness 0.0456.	Highest direct pair coverage; a candidate to test, not a universal-preference claim.
Compromise Bridge candidate	Playful-Streetwear: weighted betweenness 0.3134; Bridge Load 33.0 fractional path-load units.	Highest baseline weighted betweenness in this graph; a hypothesis for indirect style mediation.
Direct overlap	58 / 120 MBTI pairs = 48.3% have no shared direct Fashion Vibe.	This percentage describes pair overlap and is separate from Bridge Load.

Bridge Load definition. For every MBTI pair with no shared direct Vibe, the code finds weighted shortest paths, splits that pair's contribution equally across tied shortest paths, and adds the fractional contribution to each intermediate Vibe. Therefore **33.0 is a path-load quantity, not a percentage or a count of unique pairs**. The separate 48.3% figure is 58 of the 120 possible MBTI pairs with no direct shared Vibe.

SENSITIVITY & BUSINESS IMPLICATIONS

In six leave-one-creator-out runs, Playful-Streetwear remains the top bridge in 4 runs; Dark-Perfectionist leads once and Soft-Romantic leads once. The ranking is sensitive to creator composition, so the business action should be to pilot a shortlist (Playful-Streetwear plus the alternatives indicated by the sensitivity table), not hard-code one universal bridge. The stricter co-occurrence tests should be interpreted alongside graph fragmentation and unreachable pairs.

MINIMUM CO-OCCURRENCE THRESHOLD SENSITIVITY

Min. co-occurrence	Retained edges	Largest component	Unreachable pairs	Nominal top bridge (weighted BC)
1 (baseline)	51	28 nodes	0	Playful-Streetwear (0.3134)
2	14	7 nodes	111	High-Energy (0.0313)
3	2	3 nodes	119	Sharp-Tailored (0.0028)

Interpretation. At thresholds 2 and 3, the graph fragments sharply and most no-overlap pairs become unreachable. High-Energy and Sharp-Tailored are nominal winners only in these fragmented graphs and should not be promoted as business alternatives. The baseline graph is the primary exploratory result; validate a shortlist with actual couples.

Limitations. N = 67 is small and concentrated among four main creators. MBTI labels were taken from social-media content, not independently validated; Fashion Vibes and Room Style mappings are researcher-defined. The network does not establish causation, psychological validity, actual consumer preference, or relationship compatibility. A 100-couple test is a proposed future validation, not a completed experiment.

Source: repository CSV and regenerated notebook analysis. Run the staged Notebook from top to bottom to regenerate audit files, metrics, sensitivity tables, figure, and interactive output.